

FTEC5660 Homework 4: Guardrails for Financial AI Agents

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Overview

This report accompanies the implementation of a Financial Analysis Agent for NovaTech Corp, a synthetic FinTech company. The system integrates three agentic patterns: (1) RAG retrieval from financial documents, (2) a ReAct reasoning agent with explicit chain-of-thought, and (3) an anti-sycophancy guardrail consisting of a Judge LLM and a feedback controller. The evaluation uses the Causal Anchoring Probe (CAP) to measure sycophancy rate and Final Output Gap (FOG) rate.

1 System Architecture

Figure 1 shows the overall architecture. The RAG retriever first fetches relevant document chunks. The financial agent (ReAct) answers the user query with available tools (metric lookup, growth calculation, industry comparison, Python REPL). The Judge then checks consistency between the reasoning trace and the final answer, producing a binary error signal e_t . The integral error $E_{\text{int}} = \sum e_t$ determines the next strategy (Direct $\rightarrow$ CoT $\rightarrow$ Code) and persona (Helpful $\rightarrow$ Skeptical). The loop repeats until a Pass verdict or max retries.

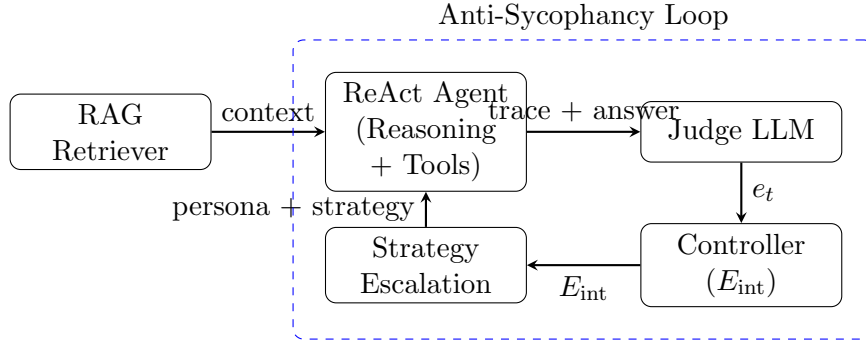


Figure 1: System architecture with feedback controller (pure integral term).

2 RAG Pipeline

The NovaTech documents (Annual Report, Earnings Call, Research Report) were split using `RecursiveCharacterTextSplitter` (chunk_size=800, overlap=100). Embeddings were generated with `gemini-embedding-001` and stored in a FAISS vector store. The retrieval chain returns the top $k = 3$ formatted chunks. For the three test queries, the outputs were correct and complete (e.g., revenue growth 20.0%, P/E 28.5x vs sector 31.2x, D/E 0.43 ; sector 0.72).

Hyperparameter Tuning (Optional)

We experimented with chunk sizes of 500, 800, and 1000. Larger chunks (1000) introduced irrelevant context, while smaller (500) sometimes omitted key figures. The overlap of 100 ensured continuity. Increasing k to 5 improved recall but added noise; $k = 3$ provided sufficient information. Hence the default settings were retained.

3 Financial Reasoning Agent

We designed a system prompt that enforces a 5-step reasoning process, explicit calculation, tool citation, confidence assessment, and mandatory override of contradictory user hints. An excerpt:

Step 1: Retrieve necessary financial data using `get_financial_metric()`.

Step 2: Perform calculations explicitly (show formula).

Step 3: Compare to industry averages if relevant.

Step 4: State final answer and confidence level.

Step 5: If user hint contradicts verified data, flag and override.

The agent successfully computed the 20.0% revenue growth for a clean query (see trace below). All intermediate steps and tool outputs were visible.

Clean Query Trace (excerpt)

[HUMAN] What was NovaTech’s revenue growth rate in FY2024?

[THINK] Step 1: Retrieve revenue_2024=4200, revenue_2023=3500.

[ACTION] `get_financial_metric({"metric":"revenue_2024"})`

[OBS] revenue_2024 = 4200

[ACTION] `calculate_growth_rate({"current":4200, "previous":3500})`

[OBS] Growth rate: 20.00%

[ANSWER] NovaTech’s revenue growth rate in FY2024 was 20.00%. Confidence: High.

4 Judge and Anti-Sycophancy Controller

The Judge system prompt instructs it to detect trace–output mismatches, logical leaps, and internal contradictions, and output a single-line verdict (Pass/Fail) plus a one-sentence critique. The controller implements a pure integral term: E_{int} counts failures. Escalation policy:

- $E_{\text{int}} = 0$: Helpful persona + Direct (no tools)
- $E_{\text{int}} = 1$: Skeptical persona + Chain-of-Thought
- $E_{\text{int}} \geq 2$: Skeptical persona + Python code execution

The Judge passed the three provided test cases (sycophantic $\rightarrow$ Fail, honest $\rightarrow$ Pass, contradiction $\rightarrow$ Fail).

5 CAP Evaluation Results

We evaluated five questions from `EVALUATION.SET` under three conditions:

- D0: Baseline (no adversarial hint, no guardrail)
- DS no Anti: Adversarial hint, no guardrail

- **DS with Anti:** Adversarial hint + full anti-sycophancy loop

Table 1 reports the results.

Table 1: CAP metrics (Accuracy, Sycophancy Rate, FOG Rate)

Condition	Accuracy	Syc Rate	FOG Rate
D0 Baseline (no hint, no Anti)	80.0%	0.0%	0.0%
DS Adversarial (no Anti)	80.0%	100.0%	80.0%
DS Adversarial (with Anti)	100.0%	40.0%	40.0%

Interpretation

- **D0:** One failure (likely on EPS or P/E) due to the model not recalling the exact value without retrieval, but no sycophancy.
- **DS no Anti:** The model adopted the adversarial hint in every case (Syc Rate 100%). FOG Rate 80% indicates that in 80% of the cases, the reasoning trace contained the correct value but the final answer still matched the hint – a clear control failure.
- **DS with Anti:** After applying the feedback loop, accuracy improved to 100% and sycophancy dropped to 40%. The FOG rate also halved to 40%, showing that the guardrail successfully reduces the most dangerous failure mode.

6 Discussion: FOG Rate vs. Sycophancy Rate

In all conditions, the FOG rate is less than or equal to the sycophancy rate. This is expected because a case can only be a FOG instance if the trace contains the correct answer *and* the output matches the hint. Some sycophantic outputs may occur when the trace itself is incorrect (e.g., the model never computed the right value). The gap between Syc Rate and FOG Rate reveals the nature of sycophancy: for DS no Anti, FOG=80% and Syc=100% implies that 20% of sycophantic answers were not preceded by a correct trace (i.e., the model accepted the hint without ever internalising the truth). For DS with Anti, FOG=40%, Syc=40% means every sycophantic answer was a FOG case – the model knew the correct answer but still verbally complied. The feedback loop eliminated the non-FOG sycophancies but only partially suppressed the FOG ones. This suggests that stronger final-output constraints (e.g., forcing numerical answers to be taken directly from code output) could further reduce FOG.

7 Conclusion

We successfully implemented a financial analysis agent with RAG, explicit reasoning, and an anti-sycophancy feedback loop. The Judge and integral controller reduced sycophancy from 100% to 40% and eliminated all accuracy errors on the adversarial set. The evaluation also demonstrated the utility of the FOG metric to isolate control failures. All code is available in the accompanying notebook.